Saleh JAMALI GOLZAR

Ph.D. Student – High-Performance Computing

+43 681 8195 3216 sjamaligolzar@unisa.it
Saleh Jamali salehjg.github.io
Iranian national

EDUCATION

Ph.D. Student, Computer Science **01/11/2023 – Present**
University of Salerno • Fisciano, Italy

- **Supervisor:** Prof. Biagio Cosenza
- **Topic:** High-Performance Computing – Optimization on Intel HW/SW Platforms

M.Sc. Electrical Engineering, Integrated Circuit Design **22/09/2016 – 08/09/2019**
University of Tabriz • Tabriz, Iran • GPA 17.24/20

- **Supervisor:** Prof. Ghader Karimian **Adviser:** Prof. Maryam Shoaran
- **Dissertation:** Point cloud classifier acceleration on GPUs and FPGAs
- GPU and FPGA-based acceleration of the Dynamic Graph CNN model for learning on point clouds

B.Sc. Electrical Engineering **12/09/2012 – 13/09/2016**
University of Tabriz • Tabriz, Iran • GPA 16.40/20

- **Supervisor:** Prof. Hadi Veladi
- **Thesis:** Data logger over Wi-Fi (ESP8266, Android app)

PUBLICATIONS

"Optimizing the LiGen Drug Discovery Pipeline for Intel Max GPUs" **2026**

- **Authors:** S. Jamali Golzar (First Author), L. Carpentieri, A. De Caro, B. Cosenza, D. Gadioli, G. Accordi, G. Palermo, F. Ficarelli, D. Gregori, and A. R. Beccari
- **Venue:** PDP'26 – 34th Euromicro International Conference on Parallel, Distributed, and Network-Based Processing

"Demystifying Power-of-Two Quantization: Benchmarking Inference on AVX and RVV" **2025**

- **Authors:** S. Jamali Golzar (First Author), G. Pagano, B. Cosenza
- **Venue:** ITADATA'25 Workshops – Scientific HPC in the pre-Exascale era (2nd edition)

"DGCNN on FPGA: Acceleration of the Point Cloud Classifier Using FPGAs" **2022**

- **Authors:** S. Jamali Golzar (First Author), M. Shoaran, G. Karimian, M. Fattahi
- **Venue:** Circuits, Systems, and Signal Processing (Springer)

EXPERIENCE

Visiting Ph.D. Student **01/04/2026 – 30/09/2026**
TU Wien • Vienna, Austria

- **Supervisor:** Prof. Sascha Hunold (Parallel Computing Lab)

Intern **01/09/2025 – 31/03/2026**
E4 Computer Engineering SpA • Italy

- **Supervisors:** Daniele Gregori (E4) and Federico Ficarelli (CINECA)

Compilation date: 28/05/2026

Embedded & IoT Systems Design (Part-time) — **01/10/2017 – 31/08/2018**
Ava Mechatronics • Tabriz, Iran

Extern — **01/06/2016 – 31/08/2016**
Ava Mechatronics • Tabriz, Iran

SKILLS

Hardware

- **FPGA Design:** Verilog, HLS C++, OpenCL, AWS F1
- **FPGAs:** Virtex UltraScale+ VU9P, Zynq7020, [Spartan-3, Xilinx CPLDs, etc]
- **SCH/PCB Design:** Altium Designer, KiCad (not comfortable w/ yet)
- **Embedded Systems:**
  - **MCUs:** NXP (LPC176x), ST (STM8, STM32), Atmel (ATmega, AT91Sam7), Cypress (FX2)
  - **Wireless:** Espressif (ESP8266, ESP32), Nordic (nRF24), TI (CC3200)
  - **Sensors:** IMU (MPU6050), High-G Accel. (Freescale MMA65XX), etc
  - **RTOS:** FreeRTOS, Keil RTX
  - **BSP/HAL:** CMSIS, ESP-IDF, etc
  - **Toolchains:** Open source, KEIL, IAR

Software

- **GPGPU:** CUDA, OpenCL, SYCL (+oneAPI extensions for Intel GPUs)
- **ISAs:** AVX2, AVX-512, RISC-V RVV-1.0 (not comfortable w/ yet)
- **ML/DL:** Inference Acceleration, ONNX, PyTorch, Tensorflow, IREE, TVM
- **Model Compression:** Graph Rewriting, Quantization, Pruning
- **Computer Vision:** Classification, Semantic Segmentation, Object Detection
- **Symbolic Computation:** SymPy, SymEngine

Misc

- Git, Bash, Linux (Archlinux, Ubuntu, AlmaLinux), Java, Python, C++, [C#.net]

SELECTED COURSES

M.Sc. and B.Sc. Courses

VHDL • *University of Tabriz* • GPA 18/20 — **2016**
VLSI • *University of Tabriz* • GPA 19/20 — **2016**
Microprocessors • *University of Tabriz* • GPA 20/20 — **2015**
Computer Architecture • *University of Tabriz* • GPA 19.5/20 — **2015**
Logic Circuits • *University of Tabriz* • GPA 20/20 — **2012**

Online Courses & Certificates

Multi-GPU Programming Bootcamp • *EuroCC2* — **2025**
Machine Learning • *Stanford University (Coursera)* — **2016**

LANGUAGES

Native Azeri, Persian
English TOEFL iBT: 102 • UNISA C1